# Assignment 1 文杰 123090612

## Question 1

1. Objective : Minimize the sum of squared residuals :

$$\min_{\beta_0} \sum_{i=1}^{n} (y_i - \beta_0)^2$$

Taking the derivative with respect to $\beta_0$ :

$$\frac{\partial}{\partial \beta_0} \sum_{i=1}^{n} (y_i - \beta_0)^2 = -2 \sum_{i=1}^{n} (y_i - \beta_0)$$

Set derivative to 0 & solve for $\beta_0$ :

$$-2 \sum_{i=1}^{n} (y_i - \beta_0) = 0 \quad \therefore \sum_{i=1}^{n} y_i - n\beta_0 = 0 \quad \therefore \beta_0 = \frac{1}{n} \sum_{i=1}^{n} y_i$$

2. Objective : Minimize the sum of squared residuals :

$$\min_{\beta_1} \sum_{i=1}^{n} (y_i - \beta_1 x_i)^2$$

Taking the derivative with respect to $\beta_1$ :

$$\frac{\partial}{\partial \beta_1} \sum_{i=1}^{n} (y_i - \beta_1 x_i)^2 = -2 \sum_{i=1}^{n} x_i (y_i - \beta_1 x_i)$$

Set derivative to 0 & solve for $\beta_1$ :

$$-2 \sum_{i=1}^{n} x_i (y_i - \beta_1 x_i) = 0 \quad \therefore \sum_{i=1}^{n} x_i y_i - \beta_1 \sum_{i=1}^{n} x_i^2 = 0 \quad \therefore \beta_1 = \frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2}$$

# HW1 Question2

## 1. Predicted sign of $\beta_1$

- The coefficient $\beta_1$ , which represents the effect of years of education on monthly wages, is expected to be **positive**.
- This is because higher levels of education typically lead to higher-paying jobs, as education improves skills, productivity, and qualifications.

## 2. Run the regression

The results of the regression of monthly wages on years of education are as follows:

| Source | SS | df | MS | Number of obs | = | 61,136 |
|---|---|---|---|---|---|---|
| | | | | F(1, 61134) | = | 8530.77 |
| Model | 3.1880e+14 | 1 | 3.1880e+14 | Prob > F | = | 0.0000 |
| Residual | 2.2846e+15 | 61,134 | 3.7371e+10 | R-squared | = | 0.1225 |
| | | | | Adj R-squared | = | 0.1224 |
| Total | 2.6034e+15 | 61,135 | 4.2585e+10 | Root MSE | = | 1.9e+05 |

| wage | Coefficient | Std. err. | t | P>\|t\| | [95% conf. | interval] |
|---|---|---|---|---|---|---|
| yeduc | 17910.03 | 193.9109 | 92.36 | 0.000 | 17529.96 | 18290.09 |
| _cons | 57405.06 | 2004.897 | 28.63 | 0.000 | 53475.45 | 61334.66 |

- Intercept ($\hat{\beta}_0$): **57405.06** (with a *standard error* of **2004.897**)
- Slope ($\hat{\beta}_1$): **17910.03** (with a *standard error* of **193.9109**)

Other key statistics:

- $R^2$ = **0.1225** (about 12.25% of the variation in wages is explained by years of education)
- Number of observations: 61,136

### Interpretation:

- The positive coefficient of $\hat{\beta}_1$ confirms my expectation that the relationship between education and wages is positive.
- Each additional year of education is associated with an increase of 17,910.03 IDR in monthly wages.

## 3. Interpretation of the parameters:

- $\hat{\beta}_0$ = 57405.06: This is the predicted monthly wage for someone with zero years of education (which could be interpreted as the baseline wage for uneducated individuals).
- $\hat{\beta}_1$ = 17910.03: For each additional year of education, an individual's monthly wage is predicted to increase by 17,910.03 IDR.

## 4. Fitted values and residuals

The results are as follows:

| Variable | Obs | Mean | Std. dev. | Min | Max |
|---|---|---|---|---|---|
| residuals | 61,136 | -.0007623 | 193313.4 | -362695.6 | 1.00e+07 |

## Interpretation of the Output:

- **Number of Observations (Obs)**: 61,136 — This means that after dropping missing values, 61,136 valid observations are used in the regression. Stata generated residuals for these observations.
- **Mean of Residuals**: -0.0007623 — The mean of the residuals is extremely close to zero (about -0.00076). In an OLS regression, the residuals should sum to zero, or be very close to zero, due to the properties of OLS estimation. This tiny deviation from zero is due to numerical precision, and this result is expected.
- **Standard Deviation (Std. dev)**: 193,313.4 — This is the standard deviation of the residuals, showing the spread of the errors around the fitted line.
- **Minimum and Maximum**: The minimum residual is -362,695.6 and the maximum residual is 10,000,000. This indicates some large errors in prediction, possibly due to outliers or a wide range of wage data.

# 5. Predicted wage for 10 years of education:

```
. display _b[_cons] + _b[yeduc]*10
236505.32
```

# 6. Variation explained by education:

The $R^2$ value of 0.1225 indicates that about **12.25%** of the variation in wages is explained by the number of years of education. This relatively low value suggests that factors other than education contribute significantly to wage determination.

## Possible reasons why education might be lower among rural residents include:

- Limited access to schools, especially higher education.
- Economic constraints, as rural families may need children to work instead of continuing their education.
- Cultural factors that may place less emphasis on formal education in rural areas.

# 7. Alternative functional forms:

## Log-Linear Model:

$$\log(wage)_i = \beta_0 + \beta_1 education_i + \mu_i$$

```
. * Log-Linear Model: log(wage) on education
. gen log_wage = log(wage)
(91,854 missing values generated)

. reg log_wage yeduc
```

| Source | SS | df | MS | | |
|---|---|---|---|---|---|
| Model | 5317.65528 | 1 | 5317.65528 | Number of obs | = 61,136 |
| Residual | 21980.5554 | 61,134 | .359547149 | F(1, 61134) | = 14789.87 |
| | | | | Prob > F | = 0.0000 |
| | | | | R-squared | = 0.1948 |
| | | | | Adj R-squared | = 0.1948 |
| Total | 27298.2107 | 61,135 | .446523442 | Root MSE | = .59962 |

| log_wage | Coefficient | Std. err. | t | P>\|t\| | [95% conf. | interval] |
|---|---|---|---|---|---|---|
| yeduc | .073147 | .0006015 | 121.61 | 0.000 | .0719681 | .0743259 |
| _cons | 11.41769 | .0062188 | 1836.01 | 0.000 | 11.4055 | 11.42988 |

- Intercept ($\hat{\beta}_0$ ): **11.41769**
- Education coefficient ($\hat{\beta}_1$): **0.073147**
- $R^2$: **0.1948**

### Interpretation :

- For each additional year of education, the wage increases by approximately 7.3147% on average.

## Linear-Log Model:

$$wage_i = \beta_0 + \beta_1 \log(education_i) + \mu_i$$

```
. * Linear-Log Model: wage on log(education)
. gen log_educ = log(yeduc)
(7,496 missing values generated)

. reg wage log_educ
```

| Source | SS | df | MS | | |
|---|---|---|---|---|---|
| Model | 2.3599e+14 | 1 | 2.3599e+14 | Number of obs | = 59,859 |
| Residual | 2.3445e+15 | 59,857 | 3.9168e+10 | F(1, 59857) | = 6024.88 |
| | | | | Prob > F | = 0.0000 |
| | | | | R-squared | = 0.0914 |
| | | | | Adj R-squared | = 0.0914 |
| Total | 2.5805e+15 | 59,858 | 4.3110e+10 | Root MSE | = 2.0e+05 |

| wage | Coefficient | Std. err. | t | P>\|t\| | [95% conf. | interval] |
|---|---|---|---|---|---|---|
| log_educ | 128514.6 | 1655.687 | 77.62 | 0.000 | 125269.5 | 131759.8 |
| _cons | -49308.2 | 3690.619 | -13.36 | 0.000 | -56541.83 | -42074.58 |

- Intercept ($\hat{\beta}_0$ ): **-49308.2**
- Log(Education) coefficient ($\hat{\beta}_1$): **128514.6**
- $R^2$: **0.0914**

### Interpretation :

- A 1% increase in education (logarithmic change) is associated with an increase in wages of approximately `1285.146` IDR.

## Log-Log Model:

$$\log(wage)_i = \beta_0 + \beta_1 \log(education_i) + \mu_i$$

```
. * Log-Log Model: log(wage) on log(education)
. reg log_wage log_educ
```

| Source | SS | df | MS | Number of obs | = | 59,859 |
|---|---|---|---|---|---|---|
| | | | | F(1, 59857) | = | 11335.32 |
| Model | 4179.58056 | 1 | 4179.58056 | Prob > F | = | 0.0000 |
| Residual | 22070.5917 | 59,857 | .368721982 | R-squared | = | 0.1592 |
| | | | | Adj R-squared | = | 0.1592 |
| Total | 26250.1723 | 59,858 | .438540751 | Root MSE | = | .60722 |

| log_wage | Coefficient | Std. err. | t | P>\|t\| | [95% conf. | interval] |
|---|---|---|---|---|---|---|
| log_educ | .5408492 | .0050799 | 106.47 | 0.000 | .5308925 | .5508059 |
| _cons | 10.95083 | .0113235 | 967.09 | 0.000 | 10.92864 | 10.97302 |

- Intercept ($\hat{\beta}_0$ ): **-10.95083**
- Log(Education) coefficient ($\hat{\beta}_1$): **0.5408492**
- $R^2$: **0.159**

### Interpretation :

- A 1% increase in years of education is associated with a 0.54% increase in wages.

## 8. SLR.4 Assumption:

**SLR.4** states that the **expected value of the error term $u$, given the independent variable $x$ (in this case, years of education), must be zero**:

$$\mathbb{E}(u|x) = 0$$

In simple terms, this assumption means that **on average**, for any given value of the independent variable (years of education), the error term (or residuals) should equal zero. This implies that **there is no systematic relationship between the independent variable and the unobserved factors** (the error term, $u$) that affect the dependent variable (wages).

### What Does SLR.4 Mean in This Case?

In this regression, we are looking at how **years of education** affects **monthly wages**. According to SLR.4, any factors that are not included in the model but still affect wages (i.e., the error term $u$) should not be correlated with education. In other words, all the unobserved influences on wages (like ability, family background, work experience, etc.) should, on average, balance out for individuals with the same level of education.

If SLR.4 holds, we can interpret the estimated coefficient $\hat{\beta}_1$ as the **causal effect** of education on wages.

### There are several reasons to question whether SLR.4 holds in this case:

#### 1. Omitted Variable Bias:

- **SLR.4 requires that there are no omitted variables** that are correlated with both years of education and wages.
- However, in reality, factors such as **ability**, **family background**, **innate talent**, and **work experience** likely affect both education and wages. If, for example, individuals with higher ability tend to get more education and also earn higher wages, the error term will be correlated with years of education, which violates SLR.4.
- This would cause **omitted variable bias**. The estimated coefficient on education $\hat{\beta}_1$ could overestimate or underestimate the true effect of education on wages if these omitted factors are important.

#### 2. Measurement Error in Education:

- If years of education are measured with error, this can also create a correlation between the error term and education, violating SLR.4. For instance, people might misreport their level of education, or there could be differences in the quality of education (which is not captured by simple years of education).

### 3. Endogeneity:

- If individuals choose their level of education based on expected future wages or other factors not included in the model (like career aspirations or family income), this introduces **endogeneity**, meaning that education is not entirely exogenous. Endogeneity violates the assumption that the error term is uncorrelated with the independent variable (years of education).

### 4. Unobserved Factors:

- Factors like **family connections**, **networking opportunities**, and **location** (rural vs. urban) might affect wages independently of education, and these factors might also correlate with education. For example, individuals in urban areas might have more educational opportunities and also have access to higher-paying jobs. If these location-related factors are not controlled for, SLR.4 may be violated.

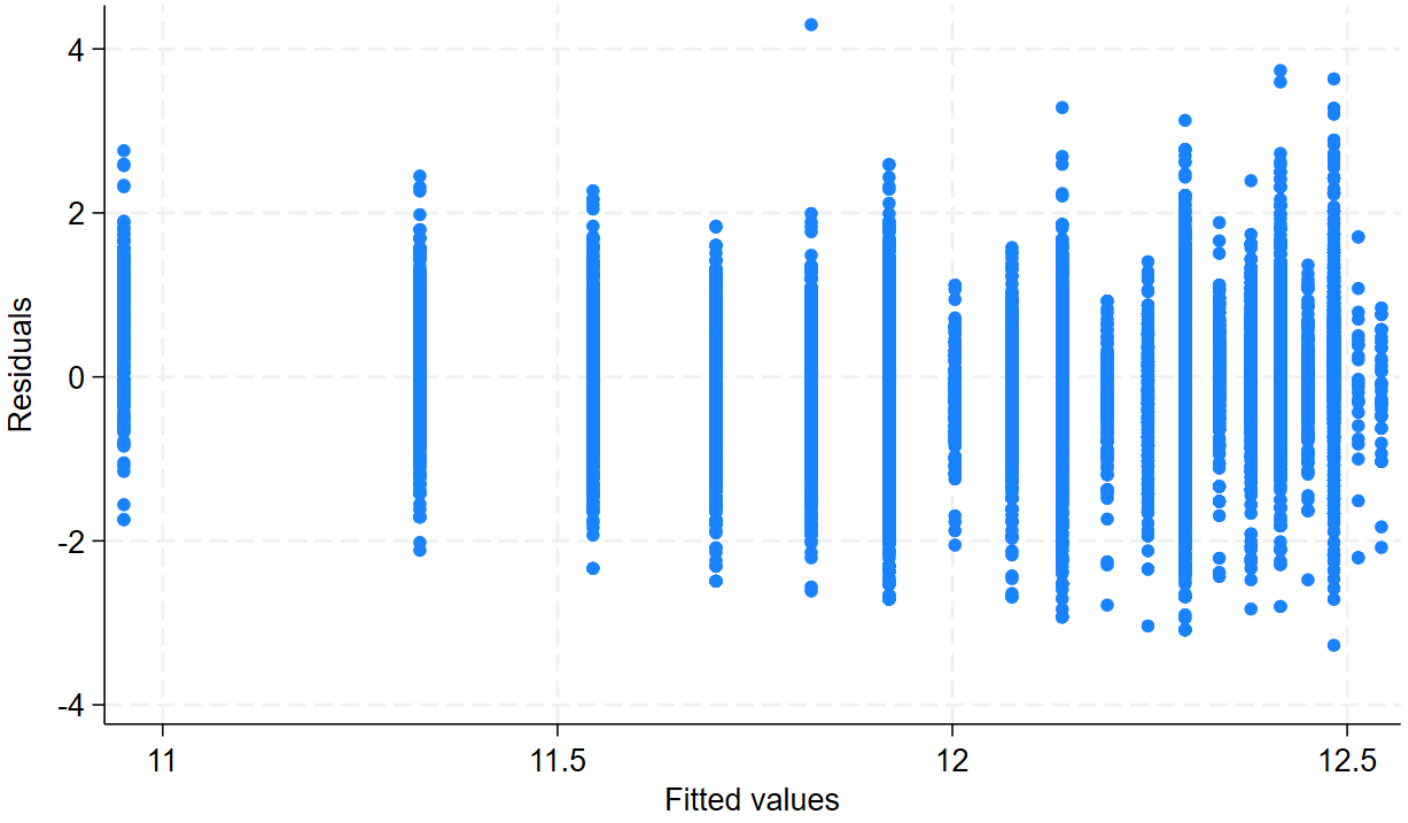
## Evidence from Residual Plot:

In order to check for violations of the SLR.4 assumption, one tool is a **residual vs. fitted values plot**. This plot helps identify any systematic patterns in the residuals:

- If the residuals are randomly scattered around zero with no clear pattern, it suggests that SLR.4 holds.
- If there are patterns (e.g., residuals systematically increase or decrease with fitted values), this indicates potential violations of SLR.4, such as heteroskedasticity or omitted variable bias.

The result is as follows:

```
. * Step 8: Plot residuals vs fitted values to check for assumption SLR.4
. rvfplot
```




## Conclusion:

It is **unlikely that SLR.4 holds perfectly** in this case due to potential **omitted variable bias** (e.g., ability, family background), **endogeneity** in education choice, and **measurement error**. Violating SLR.4 means that the estimated coefficient on education $\hat{\beta}_1$ may be biased and should not be interpreted as a purely causal effect of education on wages without additional econometric techniques, such as instrumental variable (IV) estimation, to address these issues.

## do file

- The full Stata code file with all the steps and instructions combined is as follows.

A1_Q2.do

```
* Load the data from the provided path
use "D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A1\inpresdata.dta", clear

* Step 2: Running the Simple Linear Regression: wage on years of education
reg wage yeduc

* Step 4: Compute fitted values and residuals
predict fitted_wage
predict residuals, residuals

* Verify if residuals sum to zero
summ residuals

* Step 5: Predicted wage for 10 years of education
display _b[_cons] + _b[yeduc]*10

* Step 7: Exploring alternative functional forms

* Log-Linear Model: log(wage) on education
gen log_wage = log(wage)
reg log_wage yeduc

* Linear-Log Model: wage on log(education)
gen log_educ = log(yeduc)
reg wage log_educ

* Log-Log Model: log(wage) on log(education)
reg log_wage log_educ

* Step 8: Plot residuals vs fitted values to check for assumption SLR.4
rvfplot

```

# HW1 Question3

# 1

## Answer:

The error term $\mu_i$ in the regression model $Y_i = \beta_0 + \beta_1 X_i + \mu_i$ captures all factors affecting the agricultural yield $Y_i$ that are not included in the model through the explanatory variable $X_i$ (fertilizer application). Specifically, $\mu_i$ represents the combined effect of:

- **Unobserved plot characteristics**: Factors like soil quality, microclimate variations, or terrain differences that affect yields but are not measured or included in the model.
- **Random fluctuations**: Unpredictable events such as localized pest infestations, disease outbreaks, or weather anomalies that impact individual plots differently.
- **Measurement errors**: Inaccuracies in recording yields or fertilizer application rates.
- **Variations in other inputs**: Despite efforts to keep other inputs equal, slight differences in water distribution, pesticide effectiveness, or labor efforts may occur.

## Potential Reason for Different $\mu_i$ Values:

Different plots might have varying soil fertility levels that are not accounted for in the model. For instance, one plot may have richer soil due to historical crop rotation practices, while another may have poorer soil quality. These differences affect the yield independently of the fertilizer application and result in different values of $\mu_i$ across plots.

# 2

## Answer:

Yes, $E[\mu_i|X_i] = 0$ in this case, assuming that the fertilizer quantities were truly randomly assigned to the plots and that other inputs were kept equal.

## Explanation:

- **Random Assignment**: Randomly assigning fertilizer levels ensures that any unobserved factors affecting yield (captured by $\mu_i$ are equally likely to be present in both fertilizer groups. This randomization breaks any systematic relationship between $X_i$ and $\mu_i$.
- **Equal Other Inputs**: By keeping other inputs (water, pesticides, labor) equal across plots, the farmer controls for additional variables that could correlate with both $X_i$ and $Y_i$.
- **Implication**: Under these conditions, the expected value of the error term given the fertilizer level is zero, $E[\mu_i|X_i] = 0$, satisfying one of the key assumptions of the classical linear regression model (CLRM). This assumption ensures that the estimator $\hat{\beta}_1$ is unbiased.

# 3

## (a)

### Answer:

Yes, $\hat{\beta}_1 = 0.7$ is an unbiased estimator of $\beta_1$, provided that the assumptions of the CLRM are met in this context.

### Explanation:

- **CLRM Assumptions**: One of the key assumptions is that $E[\mu_i|X_i] = 0$, which holds due to the random assignment of fertilizer quantities and equal other inputs.
- **Unbiasedness of OLS Estimators**: Under these assumptions, the Ordinary Least Squares (OLS) estimator $\hat{\beta}_1$ is unbiased, meaning $E[\hat{\beta}_1] = \beta_1$.
- **Conclusion**: Therefore, the estimated coefficient $\hat{\beta}_1 = 0.7$ reliably reflects the true effect of fertilizer application on yield, without systematic overestimation or underestimation.

## (b)

### Answer:

The assistant's concern suggests a potential violation of the CLRM assumption $E[\mu_i|X_i] = 0$ due to omitted variable bias caused by endogeneity.

### Explanation:

- **Non-Random Assignment**: If the richest land (with higher soil quality) is more likely to receive more fertilizer (40 kg/mu), then $X_i$ is correlated with soil quality.
- **Omitted Variable (Soil Quality)**: Soil quality is an unobserved variable that affects yield $Y_i$. but is not included in the regression model. It is captured in the error term $\mu_i$.
- **Endogeneity Problem**: Since $X_i$ (fertilizer quantity) is correlated with $\mu_i$ (due to soil quality), the assumption $E[\mu_i|X_i] = 0$ is violated.
- **Bias in Estimation**: This correlation leads to biased and inconsistent estimates of $\beta_1$, meaning $\hat{\beta}_1$ may not accurately represent the true effect of fertilizer on yield.

## (c)

### Answer:

#### Additional Variable to Collect:

- **Soil Quality Data** $S_i$: Collect measurements of soil quality for each plot. This could include factors like nutrient content, pH levels, organic matter, or any soil fertility indicators.

#### Regression Specifications to Investigate the Concern:

##### Test for Correlation between $X_i$ and Soil Quality $S_i$:

- **Regression Model**:

$$X_i = \gamma_0 + \gamma_1 S_i + \epsilon_i$$

- **Purpose**: To determine if fertilizer quantity assignment is correlated with soil quality.
  - **Interpretation**: A statistically significant $\gamma_1$ indicates that higher soil quality plots received more fertilizer, confirming the assistant's concern.

##### Include Soil Quality in the Yield Regression Model:

- **Regression Model**:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 S_i + \nu_i$$

- **Purpose**: To control for soil quality and assess the unbiased effect of fertilizer on yield.
- **Interpretation**:
  - If $\beta_1$ changes significantly from the original estimate $(0.7)$ after including $S_i$, it suggests that the initial estimate was biased due to omitted soil quality.
  - A significant $\beta_2$ indicates that soil quality independently affects yield.

##### Balance Test between Fertilizer Groups:

- **Comparison**:
  - Calculate the mean soil quality for plots receiving 20 kg/mu and those receiving 40 kg/mu.
- **Statistical Test**:
  - Conduct a t-test to determine if there is a significant difference in soil quality between the two groups.
- **Purpose**: To check if the randomization was compromised.

## Conclusion :

By collecting soil quality data and incorporating it into the analysis, we can verify if the assistant's concern about non-random fertilizer allocation is valid. Adjusting for soil quality allows for a more accurate estimation of the effect of fertilizer on yield and addresses potential biases arising from omitted variables.